

KENDRIYA VIDYALAYA SANGATHAN, ERNAKULAM REGION
ARTIFICIAL INTELLIGENCE (SUBJECT CODE-417), Class: X
PRE-BOARD EXAM, Session: 2024-25

Maximum Marks: 50

Time Allowed: 2 hours

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given $(5 + 16) = 21$ questions, a candidate has to answer $(5 + 10) = 14$ questions in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.

6. SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):

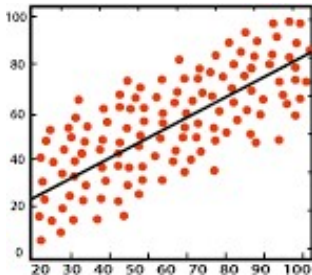
- i. This section has 05 questions.
- ii. Marks allotted are mentioned against each question/part.
- iii. There is no negative marking.
- iv. Do as per the instructions given.

7. SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):

- i. This section has 16 questions.
- ii. A candidate has to do 10 questions.
- iii. Do as per the instructions given.
- iv. Marks allotted are mentioned against each question/part.

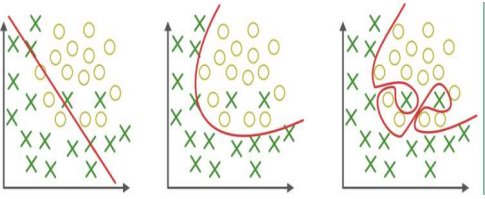
SECTION A: OBJECTIVE TYPE QUESTIONS

Q.1	Answer any 4 out of the given 6 questions on Employability Skills (1 x 4 = 4 marks)	
i.	Rohit, when leaving from his friend's house is bidding goodbye by waving his hands. Which type of non-verbal communication is used by Rohit? a) Expression b) Body language c) Gesture d) All the above	1
ii.	Software that starts working as soon as we switch on a computer is a) RAM b) ROM c) Operating System d) None of the above	1
iii.	Srishti is a young woman who makes earrings. She buys jute from a farmer and makes earrings from that. She sees that most women in her village do not work. So, she hires two women to help her. As her orders increase, she hires three more women to work for her. How was she helping her village grow? a) By selling earrings to women without a job b) By purchasing earrings from the local market c) By buying jute from the local farmer and by providing jobs to local women d) By attracting the women in her village with her creative earrings	1
iv.	Gouri gets up at 5 am and goes to her badminton classes. Then she comes home and finishes her homework before going to school. She does this all by herself. No one tells her to do it. This is an example of a) Self-motivation b) External motivation	1

	c) Both self and external motivation d) Not any specific type of motivation	
v.	Online personal journal where one can post her entries to which others can post comments is called a) Blog b) News Group c) Website d) None of the abo	1
vi.	“Efforts are made to increase the solar power generation so that our electricity needs are met and at the same time we do not pollute the environment or use up natural resources”. Which SDG can you relate this statement to? a) Life on land b) Clean water and sanitation c) Affordable and clean energy d) Reduced inequalities	1
Q.2	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	“‘This type of intelligence measure’s one’s awareness of the natural world around them and their sensitivities to any changes that occur. It allows us to identify the variation among two different species and understand how they are related’”. Identify the type of intelligence described in the above sentence	1
ii.	The Indian Government banned a few apps stating – “servers in the hostile nation are receiving and using the acquired data improperly”. Which terminology suits best for this action? a) AI Ethics b) Data Privacy c) AI Bias d) AI Access	1
iii.	Observe the graph given below:  Identify the algorithm based on the given graph (a) Dimensionality reduction (b) Classification (c) Clustering (d) Regression	1

iv.	When a user installs an app in the smartphone, it asks for access to gallery, contacts, etc. After accepting this, it gives the user agreement which most users accept without realizing the implications. What is the concern here? (a) Data Privacy (b) Unemployment (c) AI bias (d) No concern	1
v.	Search engines not only predict what popular searches may apply to your query as you start typing, but it looks at the whole picture and recognizes what you're trying to say rather than the exact search words. This is an example of (a) Computer Vision (b) Data Sciences (c) Natural Language Processing (d) Natural Language Understanding	1
vi.	A corpus contains 4 documents in which the word 'diet' was appearing once in document 1. Identify the term in which we can categorise the word 'diet'. (a) Stop word (b) Rare word (c) Frequent word (d) Removable word	1
Q.3	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Divya was learning neural networks. She understood that there were three layers in a neural network. Help her identify the layer that does processing in the neural network. (a) Output layer (b) Hidden layer (c) Input layer (d) Data layer.	1
ii	For better efficiency of an AI project Training data should be _____ i. Relevant ii. Scattered iii. Structured iv. Authentic Choose the correct option: (a) Both i and ii (b) Both i and iv (c) Only i (d) Only iv	1
iii.	F1 Score is the measure of the balance between (a) Accuracy and Precision (b) Precision and Recall (c) Recall and Accuracy (d) Recall and Reality	1
iii.	Smita is working on a project that involves over a lakh of records. Which of the following should she use to make the best project? (a) Traditional programming (b) Manual processing (c) IoT (d) Neural networks	1
iv	The _____ canvas helps you in identifying the key elements related to the problem.	1

	a) Problem scoping b) 4Ws Problem c) Project cycle d) Algorithm	
v.	The branch of AI that works with understanding language based audio inputs and producing language based audio output, is called (a) Audio Input Output Processing (b) Natural Language Processing (c) Audio Language Processing (d) Natural Language Input Output	1
vi.	Which of the following statements is not true about overfitting models? (a) This model learns the pattern and noise in the data to such extent that it harms the performance of the model on the new dataset (b) Training result is very good and the test result is poor (c) It interprets noise as patterns in the data (d) The training accuracy and test accuracy both are low	1
Q.4	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i	Which of the following is the type of data used by NLP applications? (a) Images (b) Numerical data (c) Graphical data (d) Text and Speech	1
ii	If you search Google for salons, the first few searches are mostly for female salons. This is based on the assumption that if a person is searching for a salon, in all probability that person would be female. What is this situation depicting? (a) Data Privacy (b) Data Access (c) AI Bias (d) AI Ethics	1
iii	Assertion (A): A self-driving car is made up of a combination of sensors, cameras, radar and AI Reason (R): The owner is liable if a self-driving car is involved in an accident.	1
	(a) Both A and R are correct and R is the correct explanation of A. (b) Both A and R are correct and R is not the correct explanation of A. (c) A is correct but R is incorrect. (d) A is incorrect but R is correct.	
iv	Rock, Papers, and Scissors game is based on the following domain. a) Data for AI b) Natural Language Processing c) Computer Vision d) Image Processing	1
v.	Raunak was learning the conditions that make up the confusion matrix. He came across a scenario in which the machine that was supposed to predict an animal was always predicting not an animal. What is this condition called? (a) False Positive (b) True Positive	1

	(c) False Negative (d) True Negative	
vi.	Identify the given graphs A,B,C.  a) Overfitting, Perfect, Underfitting b) Overfitting, Underfitting, Perfect c) Underfitting, Perfect, Overfitting d) Perfect, Overfitting, Underfitting	1
Q.5	Answer any 5 out of the given 6 questions (1 x 5 = 5 marks)	
i.	Aditi, a student of class XII developed a chatbot that clarifies the doubts of Economics students. She trained the software with lots of data sets catering to all difficulty levels. If any student would type or ask questions related to Economics, the software would give an instant reply. Identify the domain of AI in the given scenario. (a) Computer Vision (b) Data Science (c) Natural Language Processing (d) None of these	1
ii.	The process of extracting the meaningful root form of the word is known as _____ a) Tokenisation b) Stemming c) Lemmatisation d) Segmentation	1
iii.	Spam filtering in email is an example of _____ a) Text summarisation b) Text Classification c) Sentiment Analysis d) None of the above	1
iv.	The _____ makes the data understandable for humans as we can discover trends and patterns out of it. a) Random Data b) Graphical Representation c) Unstructured Data d) None of the above	1
v.	Deployment is a method of integrating a machine learning model into an existing production environment. (True/False)	1
vi.	Priya was confused with the terms used in the evaluation stage. Suggest her the term used for the percentage of correct predictions out of all the observations. a) Accuracy b) Precision c) Recall d) F1 Score	1

SECTION B: SUBJECTIVE TYPE QUESTIONS		
Answer any 3 out of the given 5 questions on Employability Skills (2 x 3 = 6 marks) Answer each question in 20 – 30 words.		
6	Write any two common barriers to communication and how can you address them?	2
7	Raghu always schedules all the activities he has to complete in a day. What is he doing? Elaborate on that quality you see in Raghu which you can imbibe.	2
8	What are the different types of tasks carried out in computer maintenance?	2
9	Raj has a small convenience store in his locality. There are many other convenience stores in the area. Yet, Raj's store survives the competition and does well. Which stage of an entrepreneur's career process can you relate this to? Explain.	2
10	Mention any two human activities that lead to environmental degradation.	2
Answer any 4 out of the given 6 questions in 20 – 30 words each (2 x 4 = 8 marks)		
11	How do you understand whether a machine/application is AI based or not? Explain with the help of an example	2
12	List two possible reasons for an AI model not being efficient.	2
13	If you do an image search for vacations on a popular search engine, the first few searches mostly return the picture of beaches. What is the concern here? Explain	2
14	Suhana works for a company wherein she was assigned the task of developing a project using AI project cycle. She knew that the first stage was scoping the problem. Help her list the remaining stages that she must go through to develop the project.	2
15	Explain the impact of data quality on the outcomes of data science projects, considering factors such as data completeness, accuracy, and relevance.	2
16	People of a village are totally dependent on the farmers for their daily food items. Farmers grow new seeds by checking the weather conditions every year. An AI model is being deployed in the village which predicts the chances of heavy rain to alert farmers which helps them in doing the farming at the right time. Which evaluation parameter out of precision, recall and F1 Score is best to evaluate the performance of this AI model? Explain.	2
Answer any 3 out of the given 5 questions in 50– 80 words each (4 x 3 = 12 marks)		
17	Samarth attended a seminar on Artificial Intelligence and has now been asked to write a report on his learnings from the seminar. Being a non-technical person, he understood that the AI enabled machine uses data of different formats in many of the daily based applications but failed to sync it with the right terminologies and express the details. Help Samarth define Artificial Intelligence; list the three domains of AI and the data that is used in these domains.	4
18	Akhil wants to learn how to scope the problem for an AI Project. Explain him the following: (a) 4W Problem Canvas (b) Problem Statement Template	4
19	You're explaining neural networks to your classmate Varun in a class 10 AI lesson. How would you describe the basic structure and functioning of a neural network using a simple example?	4

	Also, mention one real-world application of neural networks that you find interesting and explain how neural networks are beneficial in that context														
20	Samiksha, a student of class X was exploring the Natural Language Processing domain. She got stuck while performing the text normalisation. Help her to normalise the text on the segmented sentences given below: Document 1: Akash and Ajay are best friends. Document 2: Akash likes to play football but Ajay prefers to play online games.	4													
21	Automated trade industry has developed an AI model which predicts the selling and purchasing of automobiles. During testing, the AI model came up with the following predictions. <table><tr><th colspan="2" rowspan="2">Confusion Matrix</th><th colspan="2">Reality</th></tr><tr><th>Yes</th><th>No</th></tr><tr><th rowspan="2">Predicted</th><th>Yes</th><td>60</td><td>25</td></tr><tr><th>No</th><td>05</td><td>10</td></tr></table> (i) How many total tests have been performed in the above scenario? (ii) Calculate precision, recall and F1 Score.	Confusion Matrix		Reality		Yes	No	Predicted	Yes	60	25	No	05	10	4
Confusion Matrix				Reality											
		Yes	No												
Predicted	Yes	60	25												
	No	05	10												